

Introduction to Abstract Algebra II: Homework 6

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Victor Ostrik 10:00

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Exercise 33.2. determine whether $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis for $\mathbb{R}^3$ over $\mathbb{R}$.

Solution. The vectors $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$ are linearly independent. To see this, we can set up the equation:

$$a(1, 1, 0) + b(1, 0, 1) + c(0, 1, 1) = (0, 0, 0).$$

Writing this in matrix form, we have:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Solving this system of equations, we find that the only solution is $a = b = c = 0$. Therefore, the vectors are linearly independent.

Lastly, we show that the set of vectors spans $\mathbb{R}^3$. To do this, we reduce the matrix to row echelon form:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \leftarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Since there's a pivot in every row, the vectors span $\mathbb{R}^3$. Therefore, the set $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis for $\mathbb{R}^3$ over $\mathbb{R}$. $\square$

Exercise 33.5. Determine if $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ is a finite dimensional vector space over $\mathbb{Q}$. If it is, find a basis.

Solution. The set $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ is a finite dimensional vector space over $\mathbb{Q}$. To see this, we can express any element of the set as a linear combination of the vectors 1 and $\sqrt{2}$:

$$a + b\sqrt{2} = a \cdot 1 + b \cdot \sqrt{2}.$$

Therefore, the set is spanned by the vectors 1 and $\sqrt{2}$.

Next, we check if the vectors 1 and $\sqrt{2}$ are linearly independent. Suppose there exist rational numbers x and y such that:

$$x \cdot 1 + y \cdot \sqrt{2} = 0.$$

This implies that $x + y\sqrt{2} = 0$. Since $\sqrt{2}$ is irrational, the only way for this equation to hold is if $x = 0$ and $y = 0$. Therefore, the vectors are linearly independent.

Since the vectors 1 and $\sqrt{2}$ are linearly independent and span the set, they form a basis for the vector space. Thus, a basis for $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ over $\mathbb{Q}$ is $\{1, \sqrt{2}\}$. $\square$

Exercise 33.6. Determine if $\left\{ \frac{a+b\sqrt{3}}{c+d\sqrt{3}} \mid a, b, c, d \in \mathbb{Q}, c+d \neq \sqrt{3} \right\}$ is finite dimensional vector space over $\mathbb{Q}$. If it is, find a basis.

Solution. The set $\left\{ \frac{a+b\sqrt{3}}{c+d\sqrt{3}} \mid a, b, c, d \in \mathbb{Q}, c+d \neq \sqrt{3} \right\}$ is a finite dimensional vector space over $\mathbb{Q}$. To see this, we can express any element of the set as a linear combination of the vectors 1 and $\sqrt{3}$:

$$\frac{a + b\sqrt{3}}{c + d\sqrt{3}} = \frac{(a + b\sqrt{3})(c - d\sqrt{3})}{(c + d\sqrt{3})(c - d\sqrt{3})} = \frac{(ac - 3bd) + (bc - ad)\sqrt{3}}{c^2 - 3d^2}.$$

Therefore, the set is spanned by the vectors 1 and $\sqrt{3}$.

Next, we check if the vectors 1 and $\sqrt{3}$ are linearly independent. Suppose there exist rational numbers x and y such that:

$$x \cdot 1 + y \cdot \sqrt{3} = 0.$$

This implies that $x + y\sqrt{3} = 0$. Since $\sqrt{3}$ is irrational, the only way for this equation to hold is if $x = 0$ and $y = 0$. Therefore, the vectors are linearly independent.

Since the vectors 1 and $\sqrt{3}$ are linearly independent and span the set, they form a basis for the vector space. Thus, a basis for $\left\{ \frac{a+b\sqrt{3}}{c+d\sqrt{3}} \mid a, b, c, d \in \mathbb{Q}, c+d \neq \sqrt{3} \right\}$ over $\mathbb{Q}$ is $\{1, \sqrt{3}\}$. $\square$

Exercise 33.7. Determine if $\mathbb{R}$ is a finite dimensional vector space over $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. If it is, find a basis.

Solution. The set $\mathbb{R}$ is not a finite dimensional vector space over $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. To see this, we can consider the dimension of $\mathbb{R}$ over $\mathbb{Q}$. Since $\mathbb{R}$ is an infinite-dimensional vector space over $\mathbb{Q}$, it cannot be a finite-dimensional vector space over any subfield of $\mathbb{R}$ that contains $\mathbb{Q}$. Therefore, $\mathbb{R}$ is not a finite dimensional vector space over $\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$. $\square$

Exercise 33.8. Determine if $\mathbb{R}$ is a finite dimensional vector space over $\mathbb{C}$. If it is, find a basis.

Solution. The set $\mathbb{R}$ is not a vector space over $\mathbb{C}$. For a set V to be a vector space over a field F , it must be closed under scalar multiplication, meaning $a \cdot v \in V$ for all $a \in F$ and $v \in V$. If we take the scalar $i \in \mathbb{C}$ and the vector $1 \in \mathbb{R}$, their product is $i \cdot 1 = i$. Since $i \notin \mathbb{R}$, the closure property fails. Therefore, $\mathbb{R}$ cannot be a vector space over $\mathbb{C}$. $\square$

Exercise 33.9. Determine if $\mathbb{R}$ is a finite dimensional vector space over $\mathbb{Q}$. If it is, find a basis.

Solution. The set $\mathbb{R}$ is not a finite dimensional vector space over $\mathbb{Q}$. To see this, we can consider the dimension of $\mathbb{R}$ over $\mathbb{Q}$. Since $\mathbb{R}$ is an infinite-dimensional vector space over $\mathbb{Q}$, it cannot be a finite-dimensional vector space over $\mathbb{Q}$. Therefore, $\mathbb{R}$ is not a finite dimensional vector space over $\mathbb{Q}$. $\square$

Exercise 33.10. There is a field E with 32 elements. Determine which prime field is isomorphic with a subfield of E and determine the dimension of E over its prime field.

Solution. To determine which prime field is isomorphic with a subfield of E , we need to find the characteristic of E . The characteristic of a field is the smallest positive integer p such that $p \cdot 1 = 0$ in the field. Since E has 32 elements, it must have characteristic 2, because 32 is a power of 2. Therefore, the prime field that is isomorphic with a subfield of E is $\mathbb{F}_2$, the field with 2 elements. $\square$

Exercise 33.21. Prove that if V is a finite-dimensional vector space over a field F , then a subset $\{\beta_1, \beta_2, \dots, \beta_n\}$ of V is a basis for V over F if and only if every vector in V can be expressed uniquely as a linear combination of the β_i .

Solution. Assume that $\{\beta_1, \beta_2, \dots, \beta_n\}$ is a basis for V over F . By definition of a basis, the vectors are linearly independent and span V . Therefore, every vector in V can be expressed as a linear combination of the β_i . To show uniqueness, suppose there are two different linear combinations of the β_i that represent the same vector:

$$a_1\beta_1 + a_2\beta_2 + \dots + a_n\beta_n = b_1\beta_1 + b_2\beta_2 + \dots + b_n\beta_n.$$

Rearranging this equation gives:

$$(a_1 - b_1)\beta_1 + (a_2 - b_2)\beta_2 + \dots + (a_n - b_n)\beta_n = 0.$$

Since the β_i are linearly independent, it follows that $a_i - b_i = 0$ for all i , which means $a_i = b_i$ for all i . Thus, the representation of each vector in terms of the basis is unique.

Now, conversely, assume that every vector in V can be expressed uniquely as a linear combination of $\{\beta_1, \beta_2, \dots, \beta_n\}$. To show that these vectors form a basis, we need to show that they are linearly independent and span V .

First, we show linear independence. Suppose there exist scalars $c_1, c_2, \dots, c_n$ such that:

$$c_1\beta_1 + c_2\beta_2 + \dots + c_n\beta_n = 0.$$

Since the representation of each vector is unique, the only way for this equation to hold is if $c_i = 0$ for all i . Therefore, the vectors are linearly independent.

Next, we show that the vectors span V . Since every vector in V can be expressed as a linear combination of $\{\beta_1, \beta_2, \dots, \beta_n\}$, it follows that the vectors span V . Therefore, $\{\beta_1, \beta_2, \dots, \beta_n\}$ is a basis for V over F . $\square$

Exercise 33.23. Prove that every finite-dimensional vector space V of dimension n over a field F is isomorphic to the vector space F^n of Exercise 19.

Solution. Let V be a finite-dimensional vector space of dimension n over a field F . Since V has dimension n , there exists a basis $\{\beta_1, \beta_2, \dots, \beta_n\}$ for V over F . We can define a function $\varphi : V \rightarrow F^n$ as follows: For any vector $\alpha \in V$, express α as a linear combination of the basis vectors:

$$\alpha = a_1\beta_1 + a_2\beta_2 + \dots + a_n\beta_n.$$

Then we define:

$$\varphi(\alpha) = (a_1, a_2, \dots, a_n).$$

This function φ is well-defined because the representation of each vector in terms of the basis is unique.

Next, we need to show that φ is an isomorphism. To show that φ is linear, we check the following properties for any vectors $\alpha, \gamma \in V$ and any scalar $a \in F$:

$$\begin{aligned} \varphi(\alpha + \gamma) &= \varphi((a_1\beta_1 + a_2\beta_2 + \dots + a_n\beta_n) + (c_1\beta_1 + c_2\beta_2 + \dots + c_n\beta_n)) \\ &= (a_1 + c_1, a_2 + c_2, \dots, a_n + c_n) = (a_1, a_2, \dots, a_n) + (c_1, c_2, \dots, c_n) = \varphi(\alpha) + \varphi(\gamma), \end{aligned}$$

and

$$\varphi(a\alpha) = \varphi(a(a_1\beta_1 + a_2\beta_2 + \dots + a_n\beta_n)) = (aa_1, aa_2, \dots, aa_n) = a(a_1, a_2, \dots, a_n) = a\varphi(\alpha).$$

Therefore, φ is a linear transformation.

To show that φ is injective, suppose $\varphi(\alpha) = \varphi(\gamma)$ for some $\alpha, \gamma \in V$. This means that the coordinates of α and γ with respect to the basis are the same, which implies that $\alpha = \gamma$. Therefore, φ is injective.

To show that φ is surjective, let $(b_1, b_2, \dots, b_n) \in F^n$ be any vector. We can construct a vector $\alpha \in V$ as follows:

$$\alpha = b_1\beta_1 + b_2\beta_2 + \dots + b_n\beta_n.$$

Then we have:

$$\varphi(\alpha) = (b_1, b_2, \dots, b_n).$$

Therefore, φ is surjective.

Since φ is a linear transformation that is both injective and surjective, it is an isomorphism. Thus, V is isomorphic to F^n . $\square$

Exercise 33.24. Let V and V' be vector spaces over the same field F . A function $\varphi : V \rightarrow V'$ is a *linear transformation* of V into V' if the following conditions are satisfied for all $\alpha, \beta \in V$, and $a \in F$:

$$\begin{aligned} \varphi(\alpha + \beta) &= \varphi(\alpha) + \varphi(\beta) \\ \varphi(a\alpha) &= a\varphi(\alpha) \end{aligned}$$

- (i) If $\{\beta_i | i \in I\}$ is a basis for V over F , show that a linear transformation $\varphi : V \rightarrow V'$ is completely determined by the vectors $\varphi(\beta_i) \in V'$.
- (ii) Let $\{\beta_i | i \in I\}$ be a basis for V , and let $\{\beta'_i | i \in I\}$ be any set of vectors, not necessarily distinct, of V' . Show that there exists exactly one linear transformation $\varphi : V \rightarrow V'$ such that $\varphi(\beta_i) = \beta'_i$.

Solution to (i). Assume that $\{\beta_i | i \in I\}$ is a basis for V over F . Since $\{\beta_i | i \in I\}$ is a basis, every vector $\alpha \in V$ can be expressed as a linear combination of the basis vectors:

$$\alpha = a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_k\beta_{i_k}.$$

Applying the linear transformation φ to both sides, we have:

$$\varphi(\alpha) = \varphi(a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_k\beta_{i_k}) = a_1\varphi(\beta_{i_1}) + a_2\varphi(\beta_{i_2}) + \cdots + a_k\varphi(\beta_{i_k}).$$

Therefore, the value of $\varphi(\alpha)$ is completely determined by the values of $\varphi(\beta_i)$ for the basis vectors β_i . Since every vector in V can be expressed as a linear combination of the basis vectors, and the linear transformation is determined by its action on the basis vectors, it follows that φ is completely determined by the vectors $\varphi(\beta_i) \in V'$. $\square$

Solution to (ii). Let $\{\beta_i | i \in I\}$ be a basis for V , and let $\{\beta'_i | i \in I\}$ be any set of vectors in V' . We want to show that there exists exactly one linear transformation $\varphi : V \rightarrow V'$ such that $\varphi(\beta_i) = \beta'_i$ for all $i \in I$.

First, we define a function $\varphi : V \rightarrow V'$ as follows: For any vector $\alpha \in V$, express α as a linear combination of the basis vectors:

$$\alpha = a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_k\beta_{i_k}.$$

Then we define:

$$\varphi(\alpha) = a_1\beta'_{i_1} + a_2\beta'_{i_2} + \cdots + a_k\beta'_{i_k}.$$

This definition ensures that $\varphi(\beta_i) = \beta'_i$ for all $i \in I$.

Next, we need to show that φ is a linear transformation. For any vectors $\alpha, \gamma \in V$ and any scalar $a \in F$, we have:

$$\begin{aligned} \varphi(\alpha + \gamma) &= \varphi((a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_k\beta_{i_k}) + (c_1\beta_{j_1} + c_2\beta_{j_2} + \cdots + c_m\beta_{j_m})) \\ &= (a_1 + c_1)\beta'_{i_1} + (a_2 + c_2)\beta'_{i_2} + \cdots + (a_k + c_k)\beta'_{i_k} + (c_1)\beta'_{j_1} + (c_2)\beta'_{j_2} + \cdots + (c_m)\beta'_{j_m} \\ &= (a_1\beta'_{i_1} + a_2\beta'_{i_2} + \cdots + a_k\beta'_{i_k}) + (c_1\beta'_{j_1} + c_2\beta'_{j_2} + \cdots + c_m\beta'_{j_m}) \\ &= \varphi(\alpha) + \varphi(\gamma), \end{aligned}$$

and

$$\begin{aligned} \varphi(a\alpha) &= \varphi(a(a_1\beta_{i_1} + a_2\beta_{i_2} + \cdots + a_k\beta_{i_k})) \\ &= (aa_1)\beta'_{i_1} + (aa_2)\beta'_{i_2} + \cdots + (aa_k)\beta'_{i_k} \\ &= a(a_1\beta'_{i_1} + a_2\beta'_{i_2} + \cdots + a_k\beta'_{i_k}) \\ &= a\varphi(\alpha). \end{aligned}$$

Therefore, φ is a linear transformation. $\square$